

Adult guide — E03 The Fold Makes a Pattern

Version: planning 0.1.0 Story: The Turning-Light Trail Author: Maria Smith Status: live planning work; not an approved final publication

What this book is for

E03 is a read-aloud story and activity book for Early Years / Foundation. It continues directly from E02. The rebuilt Moon Lantern begins to show two lights, and the child helps Mira, Sol, Tavi and one new guest, Vee, repair the route to the Sunrise Arch.

The child should increasingly be able to:

1. keep two visible roles distinct without calling one positive and one

negative;

1. describe what one declared move does;
2. notice that the next lawful move brings the earlier role back;
3. copy and continue a short declared recurrence;
4. retain and identify the first move that breaks the rule; and
5. use the same two-role rule after the names and pictures change.

This is not a conventional pattern book with Fold vocabulary added. The topic and sequence come from the admitted Foundation dependency spine.

Scientific authority and exact boundary

SFT-FOUNDATION-FOLD-001

- Receipt:

sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003

- Status: model admitted, depth-independent, independently replicated
- Exact role in E03: the minimal parameter-free Fold has two complete,

disjoint, equal fibres with distinct held labels and an explicit complete return relation to the One.

The blue moon and gold sun faces translate the experience of retaining two distinct roles. They are not the proof labels and the cardboard tile is not the Fold theorem.

SFT-FOUNDATION-FOLD-DYNAMICS-001

- Receipt:

sha256:ffaa7c3740dd747aecfc858391cec140dd6c1b3c8ab219ca798fccfc1785b675

- Status: model admitted, depth-independent, independently replicated
- Exact role in E03: exact Fold dynamics retains two fibre roles, gives a

declared transition, identifies the return relation and contains the first nonstatic recurrent orbit.

The book translates only the two-role turn-and-return action. It does not teach the claim's rational part arithmetic, exact doubling-and-cast equation, fibre witnesses or uniform even divisions at child level.

What the activities do not prove

A hand turn, paper card, coloured light, over/under path or star/leaf row is a bounded demonstration. None independently derives the admitted claims. The book does not assert that every alternating pattern is an SFT Fold or that a physical flip supplies scientific authority.

The following remain excluded:

- binary digits, bit axioms or Boolean authority;
- signed or positive/negative numerical labels;
- semantic numerical zero;
- floating-point equality;
- completed infinity or an endless completed pattern;
- conventional probability or physical phase as a premise; and
- any answer selected because it is familiar from another curriculum.

Story cause-chain

Keep the mission audible: restore the turning-light trail before sunrise.

- Pages 3–5: the alternating Moon Lantern light starts the trail, the trail

stops, and Vee asks for help.

- Pages 6–8: the child experiences two distinct sides before their role names

are used independently.

- Pages 9–14: one turn changes the visible side; the next turn returns the first

side.

- Pages 15–17: the child copies the actions before meeting the word pattern.

- Pages 18–21: Sol's failed attempt stays visible; the child finds and repairs

the first broken move.

- Pages 22–27: over/under transfers the same declared two-role rule and the

child chooses the complete lawful route.

- Pages 28–31: delayed recall tests the first transition; star/leaf tests

transfer after the role names change.

- Page 32: the preserved failed attempt creates the question for E04.

If a child loses the story, restate the mission and point to the lit trail and dark trail. Do not jump straight to the next answer.

Challenge and reveal answers

| Challenge | Do before revealing | Answer shown on the reveal | |---|---|---| | Page 7 | Inspect Vee's tile and both answer pictures | Blue moon side is showing | | Page 10 | Turn a flat hand or safe card once | Gold sun shows after one turn | | Page 12 | Turn the hand or card once more | Blue moon returns | | Page 16 | Point through all four shown lights | Blue moon comes next | | Page 19 | Check each move from left to right | The fourth blue moon tile is the first break; gold sun is required | | Page 23 | Say over, under, over slowly | Under comes next | | Page 26 | Inspect where every route ends and every adjacent move | Route C is complete and follows the rule | | Page 28 | Recall the first gate before looking back | Gold sun came out | | Page 30 | Say star, leaf, star without colour cues | Leaf comes next |

Wrong answers are useful records. Ask “What is showing now?” and “What does one move do?” Preserve the chosen picture until the child has compared it. Do not say “No, that is wrong” or praise a child as clever for the answer.

Plain-language meanings

- Side: one named face of the same tile.
- Turn: one declared move that changes which named side shows.
- Return: the next lawful move brings the earlier side back.
- Pattern: a declared rule that tells us what move comes next.
- Broken rule: one visible move does not follow the declared rule.
- Role: the place a named picture has in the declared rule. The pictures may

change while the two-role rule stays the same.

Do not use held label, fibre, phase, antipode, orbit or two-cycle as unexplained child shorthand.

Safe physical activities

Use only a large card, cloth square or adult hand. Do not use small loose parts with children who may mouth them. The child may:

- draw a moon on one side of a card and a sun on the other;
- turn the card on a table rather than throwing it;
- step over and under an imaginary line with clear space and adult support; or
- point to large printed roles instead of moving physically.

Every activity has a pointing or spoken alternative. Movement is optional.

Accessibility

- Blue moon and gold sun are distinguished by symbol and name as well as colour.
- Over and under use words and route position, not colour.
- Challenge choices remain visible until the labelled reveal.

- All image descriptions name the current role, the move and the location of a broken move.
- The accessible HTML edition must preserve page order and expose every picture description beside its page text.
- The game must work by touch, keyboard and assistive technology and must honour reduced-motion settings.
- Local narration must match the visible caption exactly; sound effects are not required to understand an answer.

Companion Level Three contract

Level Three must retain Mira, Sol and Tavi as the permanent team and introduce only Vee. It must include eight or more story stages, each with a distinct, replayable mini-game:

1. catch the two lantern lights and reveal the stopped trail;
2. inspect both sides of one tile;
3. perform the first turn;
4. perform the return;
5. copy and continue the recurrence;
6. preserve, identify and repair a broken move;
7. transfer the rule to over/under roles;
8. choose the complete route; and
9. recall and transfer to star/leaf roles.

The game is not a menu of questions. Characters move in the scene, each action changes the visible trail, progress restores exactly after tab or page changes, and every completed mini-game offers replay.

Optional book codes

Codes never reveal an answer or skip a learning action.

| Code | Page | Harmless Level Three unlock | |---|---:|---| | TRAILLIGHT | 5 | the Sunrise Arch plays a short two-light greeting | | TWOSIDES | 8 | Vee's goggles reflect both role symbols | | TURNBACK | 14 | a slow-motion turn-and-return replay | | NEXTLIGHT | 17 | Sol performs a two-step light dance after completion | | KEEPTHETRY | 21 | the preserved first-try card receives a gold frame | | OVERUNDER | 24 | the bridge plays two different route notes | | RIGHTRoute | 27 | a cosmetic light trail appears after the route is solved | | NEWROLES | 31 | a small preview of E04's checking desk |

External references

The UK Early Years Foundation Stage framework may calibrate activity length, communication demands and accessibility. It is not authority for SFT claims or their dependency order. Publisher styles, conventional curricula and other games may inform

usability comparisons only.

Review and publication status

Planning version 0.1.0 remains under edu/. It may be rendered and tested as a review work. It must not be copied into publications/education, presented as approved or remotely hosted as a child service until Maria Smith explicitly approves the exact final book and companion level.